

Executive Summary

This project is a lightweight cloud-native infrastructure analytics platform built to explore and operationalize a modern data engineering stack centered around Python, BigQuery, dbt, Airflow, Cloud Run, Docker, and GCP-native services.

Originally developed as a focused three-business-day learning sprint, the project modernized an existing Bash-based operational reporting workflow into a modular, orchestrated analytics platform capable of ingesting, transforming, and exposing infrastructure resource data across multiple GCP projects.

The platform ingests Compute Engine instance, disk, and snapshot metadata directly from GCP APIs using custom Python ingestion services deployed as Cloud Run jobs. Raw operational data is loaded into partitioned BigQuery tables and transformed using dbt staging and mart layers into analytics-ready datasets supporting infrastructure visibility, cost analysis, governance checks, and operational reporting.

Workflow orchestration is managed through a Dockerized Airflow environment coordinating ingestion execution, dbt transformations, and scheduled processing pipelines. Supporting platform features include structured logging, modular ingestion design, deployment automation via Cloud Build, schema evolution support, dbt seeds/tests, and lightweight operational dashboarding.

The primary objective of the project was not to build a feature-heavy analytics application, but rather to rapidly gain hands-on experience with modern warehouse-native data platform tooling and cloud-native orchestration patterns while applying realistic infrastructure operations use cases.

Core concepts explored include:

- modular ingestion framework design
- cloud-native workload execution
- workflow orchestration
- warehouse-native transformations
- infrastructure analytics modeling
- deployment automation
- operational observability
- scalable data pipeline architecture

The resulting platform demonstrates an end-to-end analytics engineering workflow spanning ingestion, warehousing, orchestration, transformation, deployment, and dashboard consumption within a cohesive modern data stack architecture.